

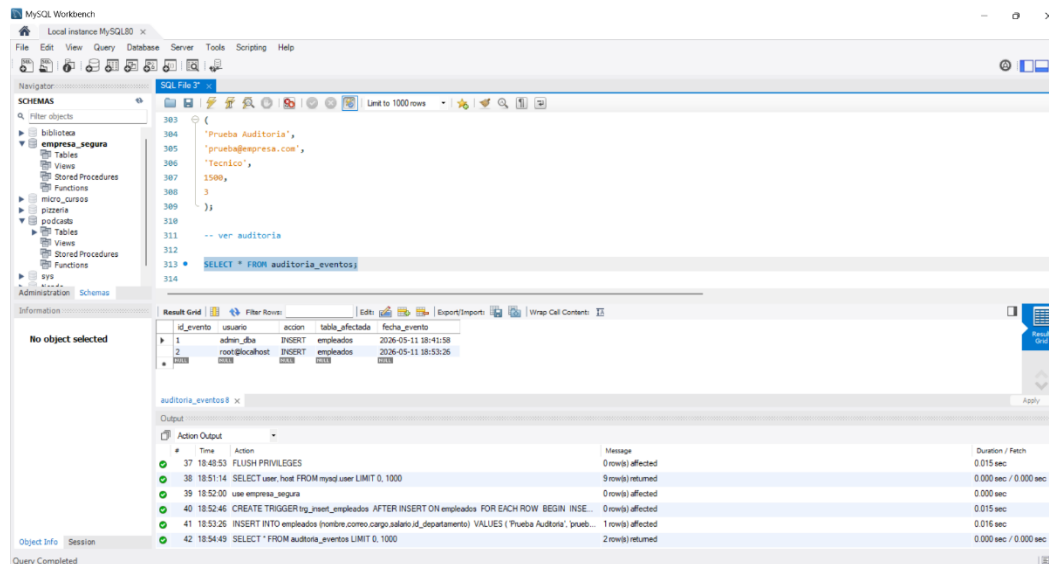
UNIVERSIDAD TECNOLÓGICA DE PANAMÁ

FACULTAD DE INGENIERÍA EN SISTEMAS COMPUTACIONALES

CURSO: BASE DE DATOS SEGURAS

NOMBRE: ISIS REYES CEDULA: 2-744-44 PROFESOR: ANTONIO CABALLERO

1. Ejecución de consultas SQL en MySQL Workbench para la creación y administración de tablas, vistas y procedimientos almacenados:



The screenshot shows the MySQL Workbench interface with the following SQL code in the editor:

```

303 (
304     "Prueba Auditoria",
305     "prueba@empresa.com",
306     "Tecnico",
307     1500,
308     3
309 );
310
311 -- ver auditoria
312
313 SELECT * FROM auditoria_eventos;
314

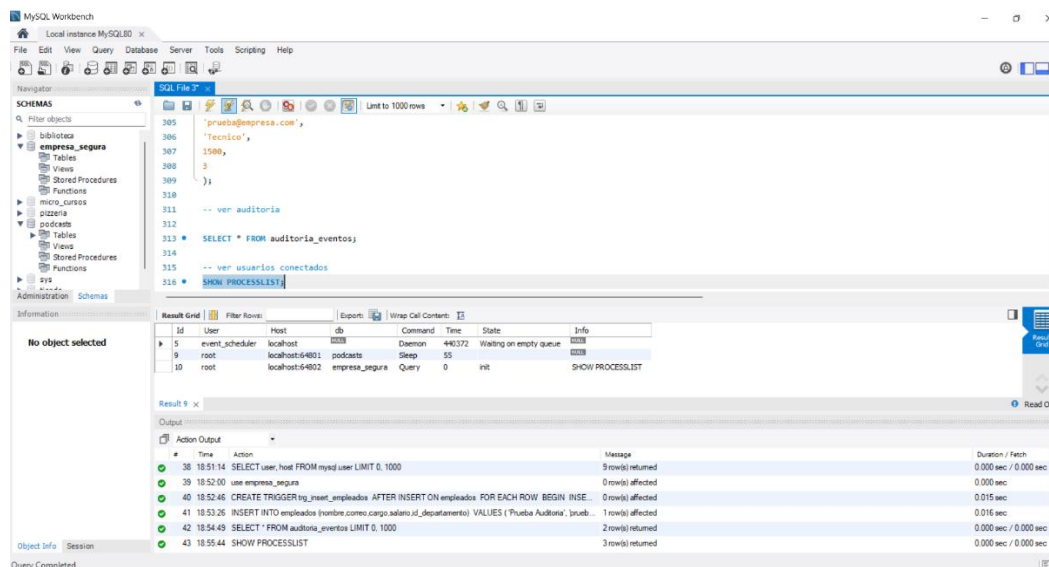
```

The Result Grid shows the following data:

id_evento	usuario	accion	tabla_afectada	fecha_evento
1	admin_iba	INSERT	empleados	2026-05-11 18:41:58
2	root@localhost	INSERT	empleados	2026-05-11 18:53:26

The Output pane shows the following messages:

#	Time	Action	Message	Duration / Fetch
37	18:48:53	FLUSH PRIVILEGES	0 row(s) affected	0.015 sec
38	18:51:14	SELECT user, host FROM mysql.user LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
39	18:52:00	use empresa_segura	0 row(s) affected	0.000 sec
40	18:52:46	CREATE TRIGGER trg_insert_empleados AFTER INSERT ON empleados FOR EACH ROW BEGIN INSE...	0 row(s) affected	0.015 sec
41	18:53:26	INSERT INTO empleados (nombre, correo, cargo, salario, id_departamento) VALUES ('Prueba Auditoria', 'prueb...	1 row(s) affected	0.016 sec
42	18:54:49	SELECT * FROM auditoria_eventos LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec



The screenshot shows the MySQL Workbench interface with the following SQL code in the editor:

```

305     "prueba@empresa.com",
306     "Tecnico",
307     1500,
308     3
309 );
310
311 -- ver auditoria
312
313 SELECT * FROM auditoria_eventos;
314
315 -- ver usuarios conectados
316 SHOW PROCESSLIST;

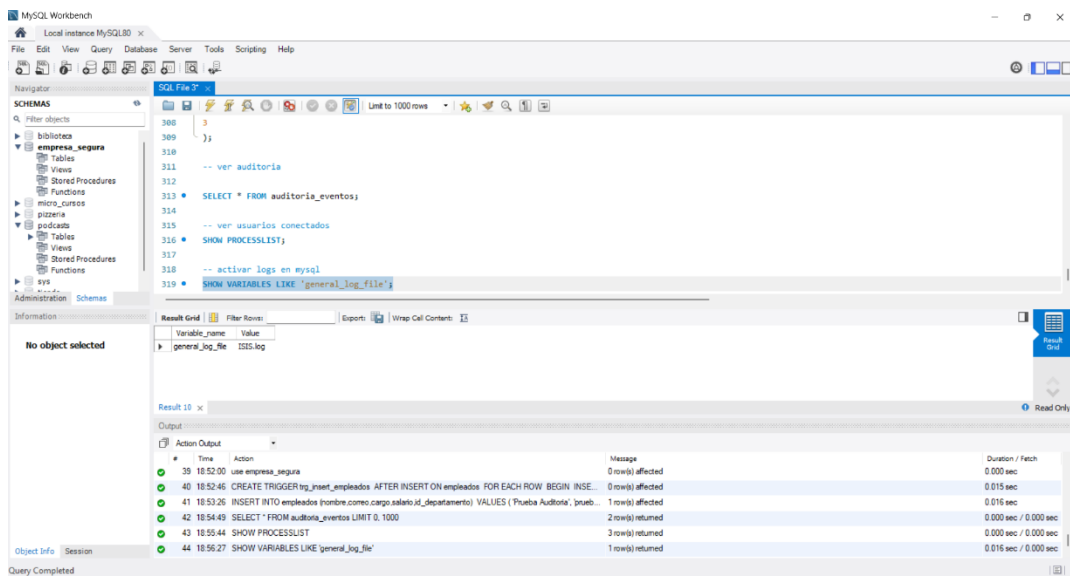
```

The Result Grid shows the following data:

ID	User	Host	db	Command	Time	State	Info
5	event_scheduler	localhost		Daemon	440/372	Waiting on empty queue	
9	root	localhost:4801	podcasts	Sleep	55		
10	root	localhost:4802	empresa_segura	Query	0	init	SHOW PROCESSLIST

The Output pane shows the following messages:

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42	18:54:49	SELECT * FROM auditoria_eventos LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
43	18:55:44	SHOW PROCESSLIST	3 row(s) returned	0.000 sec / 0.000 sec



2. Para la visualización de datos se utilizó Metabase, una herramienta de análisis y generación de dashboards:



Downloading the Open Source metabase.jar...

Your download should begin automatically. If it doesn't, you can [start it manually](#).

3. Ejecución de Metabase con Java:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\IsisR> java -version
openjdk version "25.0.3" 2026-04-21 LTS
OpenJDK Runtime Environment Temurin-25.0.3+9 (build 25.0.3+9-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.3+9 (build 25.0.3+9-LTS, mixed mode, sharing)
PS C:\Users\IsisR> cd C:\Users\IsisR\Downloads
PS C:\Users\IsisR\Downloads> dir
```

```
Windows PowerShell

PS C:\Users\IsisR\Downloads> java -jar metabase.jar
2026-05-11 19:31:59,740 INFO metabase.util :: Maximum memory available to JVM: 3.8 GB
2026-05-11 19:32:02,296 INFO util.encryption :: Saved credentials encryption is DISABLED for this Metabase instance.
For more information, see https://metabase.com/docs/latest/operations-guide/encrypting-database-details-at-rest.html
SLF4J(W): Class path contains multiple SLF4J providers.
SLF4J(W): Found provider [org.apache.logging.slf4j.SLF4JServiceProvider@bbb39957]
SLF4J(W): Found provider [org.slf4j.jul.JULServiceProvider@4c57051a]
SLF4J(W): See https://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J(I): Actual provider is of type [org.apache.logging.slf4j.SLF4JSe
```

4. Conexión de MySQL con Metabase:

3

Añade tus datos

¿Listo para empezar a explorar tus datos? Agrégalos abajo (opcional).

MySQL

Connection string (optional)

p. ej., jdbc:mysql://user:pass@host:3306/dbname?ssl=true

Puedes usar una cadena de conexión para rellenar previamente los detalles de abajo.

Nombre para mostrar

Auditoria

Host

localhost

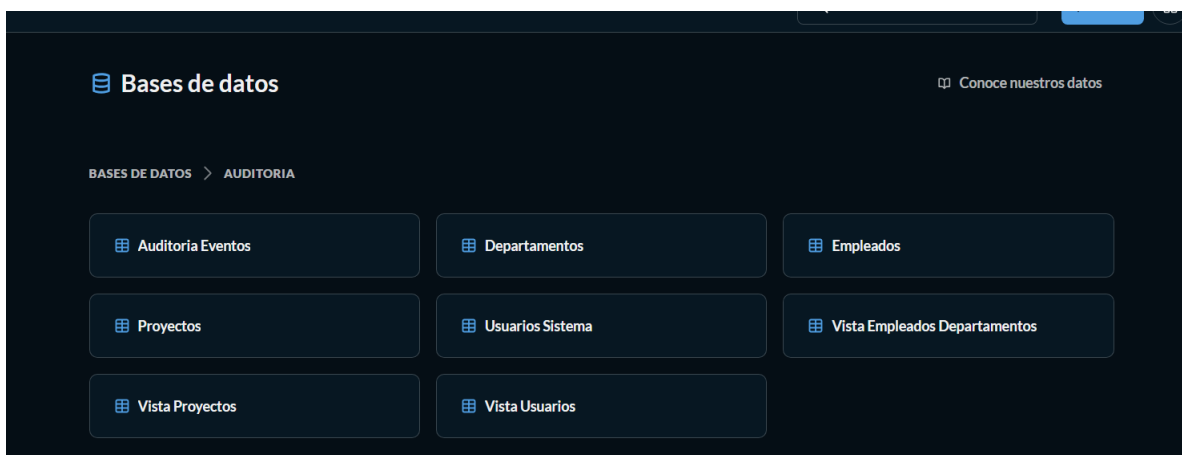
Puerto

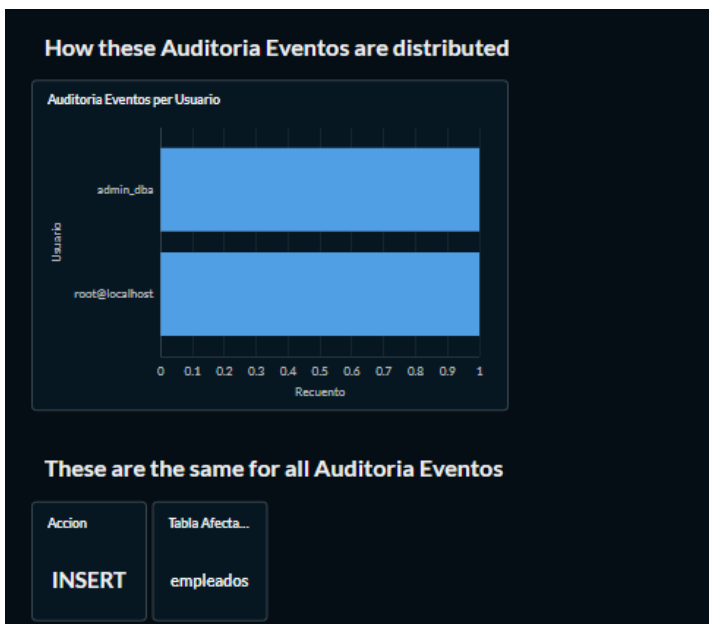
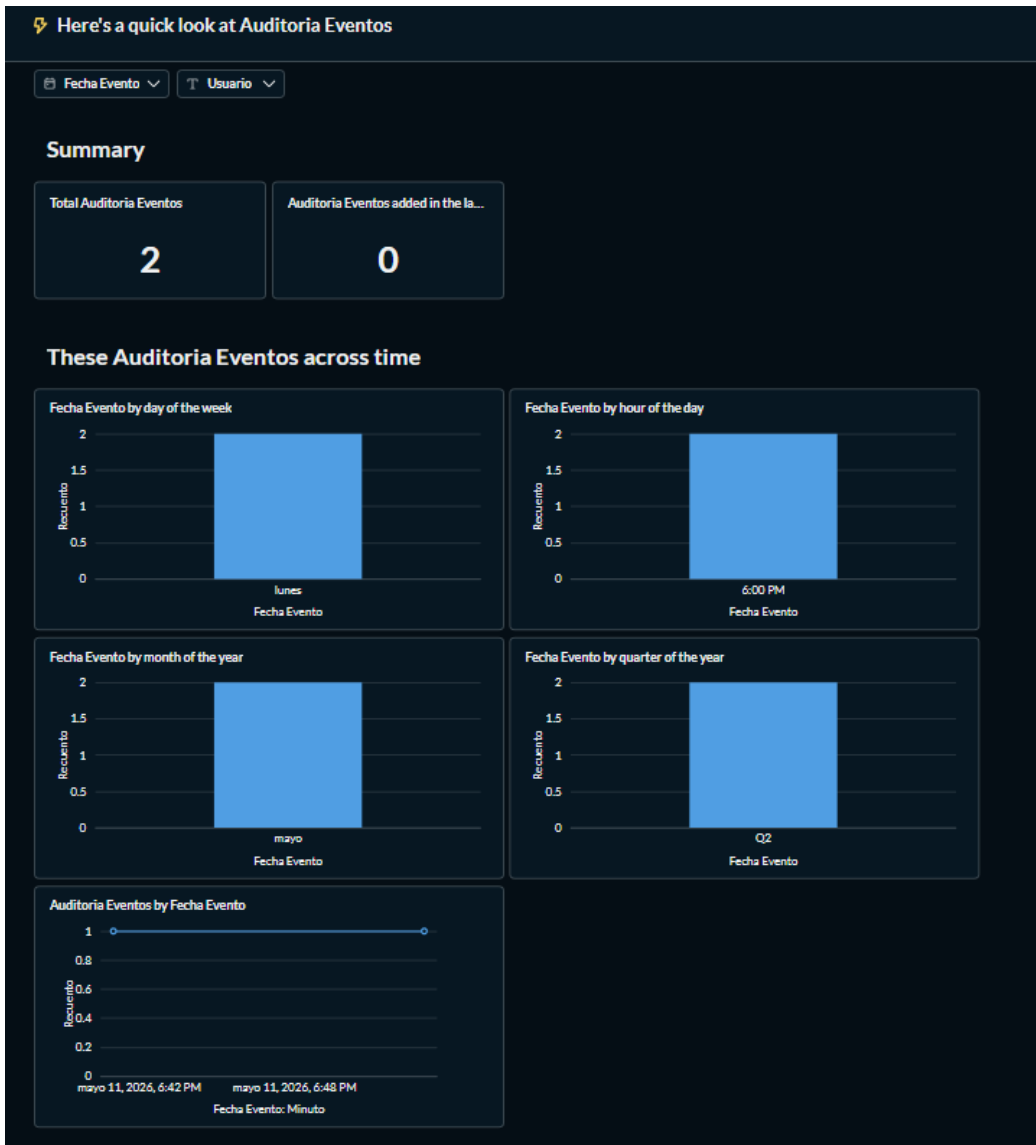
3306

Omitir

Conectar base de datos

5. Visualización de datos y generación de gráficos:





Here's a quick look at Proyectos

Fecha Inicio Nombre Proyecto Nombre

Summary

Total Proyectos

5

Proyectos added in the last 30 days

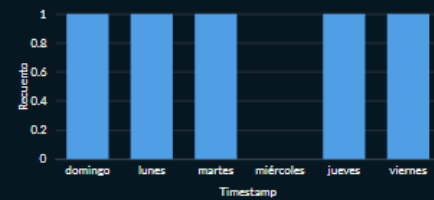
2

Distinct ID Departamento

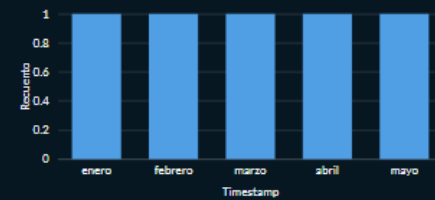
5

These Proyectos across time

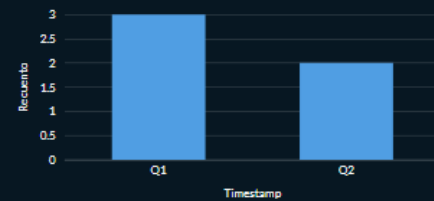
Timestamp by day of the week



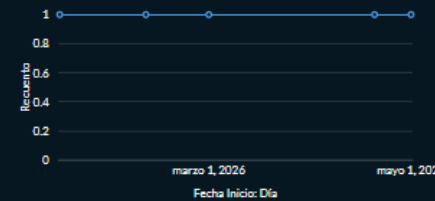
Timestamp by month of the year



Timestamp by quarter of the year



Proyectos by Timestamp



Presupuesto over time

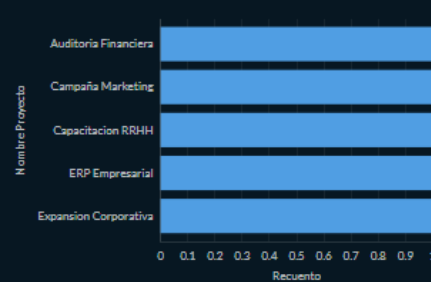


How these Proyectos are distributed

Proyectos by Presupuesto



Proyectos per Nombre Proyecto



Proyectos per ID Departamento → Nombre



Here's a quick look at Vista Proyectos

Fecha Inicio Nombre Proyecto

Summary

Total Vista Proyectos

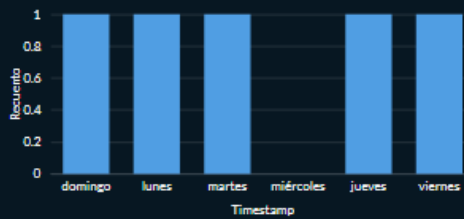
5

Vista Proyectos added in the last ...

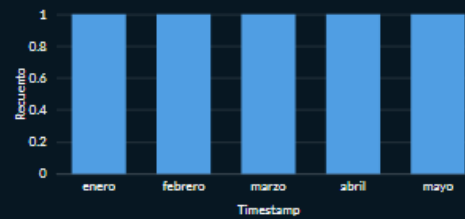
2

These Vista Proyectos across time

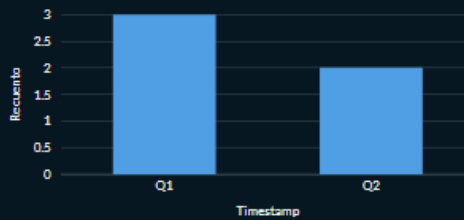
Timestamp by day of the week



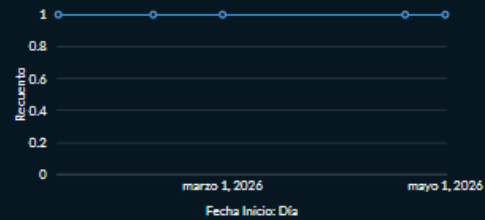
Timestamp by month of the year



Timestamp by quarter of the year



Vista Proyectos by Timestamp

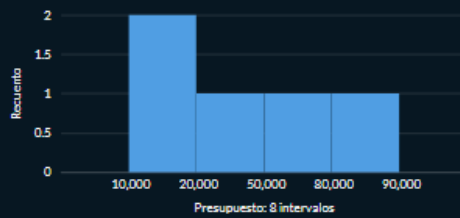


Presupuesto over time

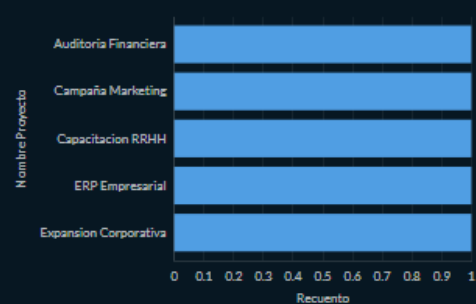


How these Vista Proyectos are distributed

Vista Proyectos by Presupuesto



Vista Proyectos per Nombre Proyecto



Here's a quick look at Departamentos

T Nombre ▾

Summary

Total Departamentos

5

How these Departamentos are distributed



Here's a quick look at Usuarios Sistema

T Usuario ▾

T Rol ▾

Summary

Total Usuarios Sistema

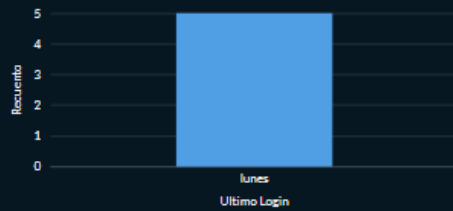
5

Usuarios Sistema added in the last...

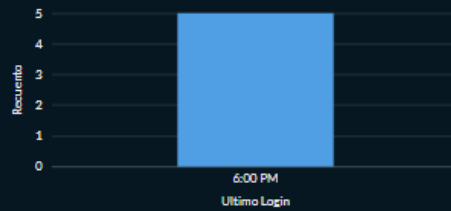
0

These Usuarios Sistema across time

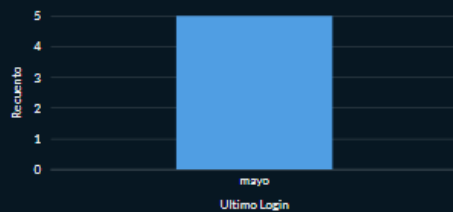
Ultimo Login by day of the week



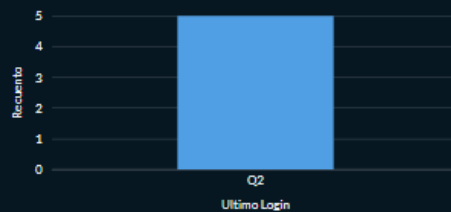
Ultimo Login by hour of the day



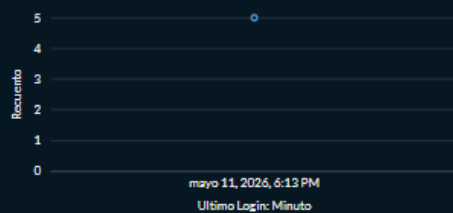
Ultimo Login by month of the year



Ultimo Login by quarter of the year

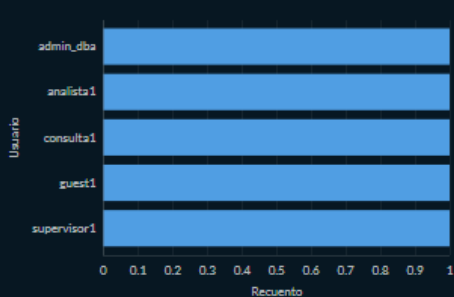


Usuarios Sistema by Ultimo Login

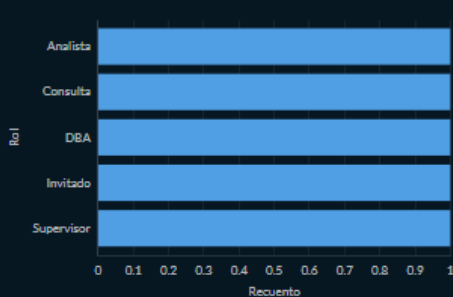


How these Usuarios Sistema are distributed

Usuarios Sistema per Usuario



Usuarios Sistema per Rol



Here's a quick look at Vista Usuarios

T Usuario ▼ T Rol ▼

Summary

Total Vista Usuarios

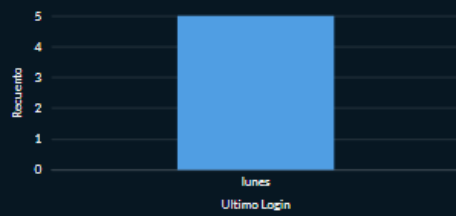
5

Vista Usuarios added in the last 3...

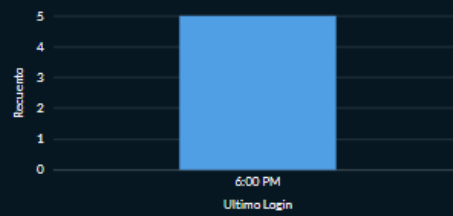
0

These Vista Usuarios across time

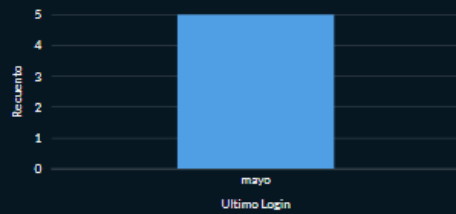
Ultimo Login by day of the week



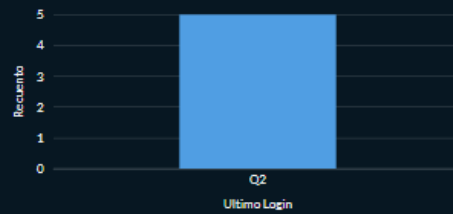
Ultimo Login by hour of the day



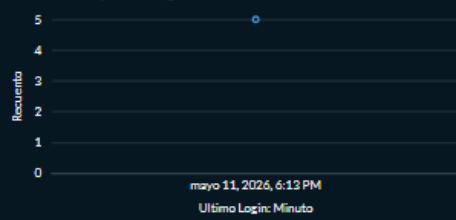
Ultimo Login by month of the year



Ultimo Login by quarter of the year

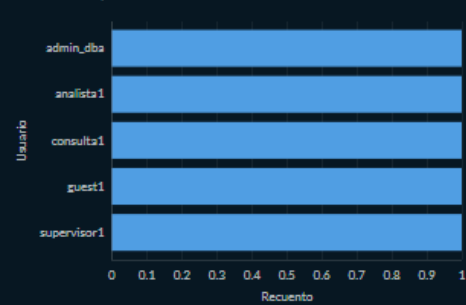


Vista Usuarios by Ultimo Login

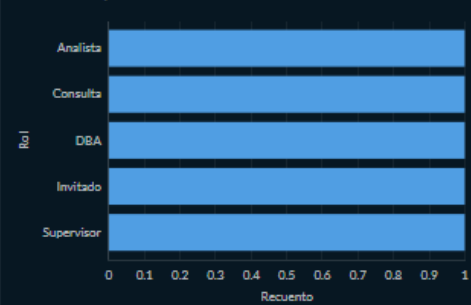


How these Vista Usuarios are distributed

Vista Usuarios per Usuario



Vista Usuarios per Rol



Here's a quick look at Empleados

T Nombre

Summary

Total Empleados

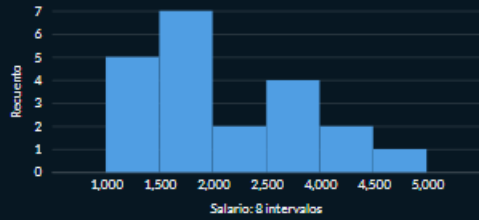
21

Distinct ID Departamento

5

How these Empleados are distributed

Empleados by Salario



Empleados per ID Departamento → Nombre



Here's a quick look at Vista Empleados Departamentos

T Departamento

Summary

Total Vista Empleados Departame...

21

How these Vista Empleados Departamentos are distributed

Vista Empleados Departamentos by Salario



Vista Empleados Departamentos per Departamento

